AI ENGINEER INTERNSHIP • TECHNICAL CASE STUDY

Intelligent Document Extraction, Validation & API Platform

AI-powered financial document extraction, validation, persistence and API platform

Supported Document Types:
Invoice • Balance Sheet • Profit & Loss Statement • Cash Flow Statement
Formats Supported: PDF (Native Digital & Scanned/Image-based), JPG, PNG | Multi-page up to 3 pages per document

Submission Component: Working Deployed Solution + Public GitHub Repository + Documentation
Primary Tech Stack: FastAPI, Google Gemini 3.6 Flash (Multimodal), PyMuPDF, SQLAlchemy, SQLite, Jinja2, Docker
Core Focus: Extraction Accuracy, API Standards, Strict Mathematical Validation, Complete Test Coverage

AI ENGINEER INTERNSHIP TECHNICAL SUBMISSION • PRODUCTION READY

PROBLEM STATEMENT & OBJECTIVES

Problem Statement & System Objectives

Platform Functional Requirements
• Dual Ingestion: Accept financial documents through both an interactive Web Frontend and automated REST APIs.
• Multi-Format Ingestion: Support PDF, JPG, and PNG files with up to 3 pages per document.
• Scanned & Digital Docs: Support native digital PDFs as well as scanned/image-based documents transparently.
• Comprehensive Field Capture: Extract all meaningful visible fields, tables, line items, and multi-period financial schedules.
• Standardized Contract: Return a deterministic, consistent, structured JSON contract with confidence & source quotes.

Integrity, Auditability & Deployment
• Financial Validation: Automatically validate document-specific arithmetic (line totals, parity, profit, cash reconciliation).
• Persistence: Store all processed results in a queryable relational database with upsert capabilities.
• API Retrieval & Dashboard: Provide indexed retrieval via REST endpoints and a complete web
dashboard view.
• Live Cloud Deployment: Containerized Docker solution deployed globally with health check monitoring.
• Zero Hallucination Rule:
"Missing information must be represented as null. The system must not invent or infer values."

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SYSTEM ARCHITECTURE

Modular Multi-Tier Solution Architecture

User Interaction
Web Browser /

Client →

Frontend UI
HTML5 / CSS / JS →

FastAPI Layer
REST APIs & Docs →

Document Engine
Validation & OCR →

AI Multimodal
Gemini 3.6 Flash →

Validation Engine
Math Reconciliation →

Persistence
SQLite /
SQLAlchemy

Frontend Tier

• HTML5, CSS3, JavaScript (Fetch API)
• Drag & drop document upload
• Document type selector (4 types)
• Dashboard table with status badges
• Comprehensive result viewer
• Raw JSON view toggle

Backend Tier

• FastAPI asynchronous framework
• 4 modular RESTful endpoints
• Automatic Swagger/OpenAPI docs
• Pydantic v2 validation models
• Structured logging & error handler
• CORS & Multipart streaming

Processing Pipeline

• File Validator: MIME, size, pages
• OCR: PyMuPDF text & 200 DPI images
• AI Engine: Gemini 3.6 Flash multimodal
• Validation: Arithmetic reconciliation
• Validation runs independently from AI

Persistence Tier

• SQLAlchemy 2.0 ORM
• Embedded SQLite database
• Indexed document_name column
• Full structured result_json storage
• Upsert-by-name latest semantics
• Automated schema initialization

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EXECUTION LIFECYCLE

10-Step End-to-End Processing Pipeline

1. Upload
User submits document (PDF/JPG/PNG) & selects document type via Web UI or POST /api/v1/documents/process.
2. Document Validation
Validates MIME type, file integrity, readability, and enforces maximum page limit (≤ 3 pages).
3. Text Extraction / OCR
PyMuPDF extracts digital text and renders pages to 200 DPI images for multimodal inspection.
4. AI Field & Table Extraction
Gemini 3.6 Flash extracts header fields, line items, and multi-period financial tables.
5. Structured JSON Generation
Strict JSON schema enforces exact typing, null values for missing data, and evidence tracking.

6. Financial Calculation Validation
Independent validation engine verifies mathematical relationships and variances.
7. Evidence & Confidence
Extracts confidence scores (0.0–1.0), source text snippets, and page numbers per field.
8. Store Result
Saves full JSON payload and queryable metadata in SQLite database via SQLAlchemy.
9. PASS / FAILED Status
Assigns overall document status (PASS if readable and extracted, FAILED on corruption/format
error).
10. Dashboard & API Delivery
Returns response instantly to caller and updates dashboard table with latest record.

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SUPPORTED DOCUMENTS

Supported Financial Documents & Extraction Schema

Document Type Extracted Fields & Structural Components Mandatory Financial Validation Checks

Invoice
Retail Receipts & B2B Tax Invoices

Invoice number, date, vendor name/address/tax ID, customer name/address/tax ID, currency, subtotal, tax rate, tax amount, discount, total amount, cash paid, change, rounding adjustment, and line items (Qty, Unit Price, Amount, VAT%).

• Quantity × Unit Price ≈ Item Amount
• sum(line_item_amounts) ≈ Subtotal
• Subtotal + Tax - Discount + Rounding ≈ Total
• Cash Paid - Total Amount ≈ Change

Balance Sheet
Consolidated Bank Statements

Company name, report title, as-of date, currency, reporting unit, auditor notes, and comparative periods containing Capital & Liabilities items + schedule refs and Asset items + schedule refs.

• Assets ≈ Liabilities + Equity
(Total Assets ≈ Total Capital & Liabilities)
• sum(Asset Components) ≈ Total Assets
• sum(C&L Components) ≈ Total C&L

Profit & Loss
Income Statement

Company name, period, currency, revenue schedules (Interest earned, Other income), expenditures (Interest expended, Operating expenses, Provisions), net profit before/after minority interest, appropriations.

• sum(Income items) ≈ Total Income
• sum(Expenditure items) ≈ Total Expenditure
• Total Income - Total Expenditure ≈ Net Profit
• Group Profit & Appropriation Balance checks

Cash Flow Statement
Multi-Page Statements

Operating activities, investing activities, financing activities, foreign exchange translation adjustments, net increase/decrease in cash, opening cash, cash acquired on amalgamation, closing cash balance.

• Operating + Investing + Financing + FX ≈ Net Change in Cash
• Opening Cash + Net Change + Adjustments ≈ Closing Cash

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REST API SPECIFICATION

REST API Design & JSON Response Contract

API Endpoints Specification

POST /api/v1/documents/process
Uploads & processes financial document. Form: file (binary) + document_type (invoice, balance_sheet, profit_and_loss, cash_flow_statement).
GET /api/v1/documents/{document_name}
Retrieves the latest structured result, extraction data, and validation audits for a specific document name.
GET /api/v1/documents
Returns summary index of all processed documents (status, confidence, timestamp) to populate dashboard.
GET /api/v1/health
Operational health check probe returning service status.

Interactive OpenAPI Documentation:
FastAPI provides automatic interactive Swagger documentation at /docs and ReDoc at /redoc.

Standard Structured JSON Contract

{
"document_name": "invoice_101.pdf",
"document_type": "invoice",
"processing_status": "PASS",
"overall_confidence": 0.98,
"file_validation": {
"file_type": "application/pdf",
"is_supported": true, "is_readable": true,
"page_count": 1, "status": "PASS"
},
"extracted_data": {
"invoice_number": {"value": "INV-2024", "confidence": 0.99}, "total_amount": {"value": 138.90, "confidence": 0.98}
},
"validation": {
"checks": [{"name": "total_check", "status": "PASS"}],
"overall_status": "PASS", "issues": []
},
"processing_metadata": {"ocr_used": false, "time_ms": 1420} }

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FINANCIAL AUDITING ENGINE

Financial Validation Engine & Status Evaluation

Independence of Extraction & Validation

The financial validation engine runs as an independent deterministic verification layer after
extraction. AI extracts values; Python arithmetic reconciles formulas.
Validation Check Schema
Every check outputs an audit record containing:
• name: Check identifier (e.g. invoice_total_check)
• formula: Human-readable formula audited
• operands: Input variables evaluated
• calculated_value: Result computed by Python
• reported_value: Number printed on document
• variance: Absolute difference |calculated - reported|
• status: PASS | FAIL | NOT_APPLICABLE

Status Rules & Tolerance Handling

PASS Criteria:
Variance is within the allowed rounding tolerance margin (±1.0). Handles minor floating-point or currency rounding differences.
FAIL Criteria:
Variance exceeds allowed tolerance, or line totals do not reconcile with subtotal. Highlighted in red in the UI.

NOT_APPLICABLE Rule:
If a required field is not present in the document (e.g. cash paid on a credit invoice), the check returns NOT_APPLICABLE.
Missing fields are never guessed, inferred, or assumed.

Core Verification Formulas: [Invoice]: Qty × Price ≈ Amount | sum(Lines) ≈ Subtotal | Subtotal + Tax - Disc ≈ Total | Cash - Total ≈ Change
[Balance Sheet]: Total Assets ≈ Total Capital & Liabilities | [P&L]: Income - Expenses ≈ Net Profit | [Cash Flow]: Op + Inv + Fin + FX ≈ Net Change | Open + Change ≈ Close

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FULL-STACK IMPLEMENTATION

Frontend Dashboard, Persistence & Cloud Deployment

Frontend Web Dashboard

• Document Ingestion: Dropdown document type selection + drag-and-drop file upload.
• Execution Feedback: Animated progress bar and status indicator during processing.
• Documents Table: View all processed files, document types, status badges, confidence %,
and timestamps.
• Result Detail View: Clean key-value grid for extracted fields with confidence color badges (High/Med/Low).
• Table Visualizer: Tabular line-item breakdown & comparative period schedules.
• Audit Breakdown: Visual validation checks table highlighting PASS / FAIL / NA.
• Raw JSON Toggle: Formatted JSON view for debugging.

Persistence, Cloud & Security

Relational Persistence:
• SQLite relational database managed via SQLAlchemy ORM.
• Stores document_name, document_type, status, confidence, and complete result JSON payload.
• Upsert semantics: re-processing same document updates record.

Production Cloud Deployment:
• Dockerized: Hermetic python:3.12-slim container.
• Render Cloud: Auto-deployed web service with dynamic port binding ($PORT).
• Public URL: Globally accessible frontend and REST API.

Security Architecture:
• API keys strictly managed via environment variables (.env).
• .gitignore protects keys and local database files from GitHub.
• Controlled error handling: zero raw stack traces exposed.

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VERIFICATION & QUALITY ASSURANCE

Testing Suite & Software Engineering Rigor

Test Suite Category Automated Test Scenarios Evaluated Engineering Outcome

API Processing Flow
test_api.py (9 tests)

• Health check endpoint (200 OK)
• Document listing index endpoint
• Unsupported file extension rejection (400 Bad Request)
• Invalid document_type rejection (422 Unprocessable Entity)
• Empty file rejection (400 Bad Request)
• Corrupted PDF handling (graceful FAILED status)
• Non-existent document 404 handling
• Dashboard & detail page HTML rendering

100% Pass Rate

Rigorous HTTP status code compliance and exception
isolation.

Financial Calculations
test_validation.py (13 tests)

• Invoice line item multiplication (Qty × Price)
• Line item subtotal summation check
• Grand total tax & discount calculation (Pass & Fail cases)
• Cash & change point-of-sale math
• Balance sheet Assets = Liabilities parity
• P&L income & expenditure component sums and net profit derivation
• Cash flow activity sums and opening→closing cash reconciliation
• NOT_APPLICABLE status handling for omitted optional fields

100% Pass Rate
Accurate validation across all 4 financial document
accounting rules.

Normalization Helpers
test_extraction.py (15 tests)

• safe_float parenthesized negatives: (500.00) → -500.00
• String numbers with comma formatting: "1,234,567.89" → 1234567.89
• None and malformed string handling (clean null conversion)
• get_field_value: extraction of scalars and nested value dictionaries
• get_nested_value: deep path traversal without KeyError exceptions

100% Pass Rate

Zero runtime crashes from unexpected OCR character
formatting.

Total Test Suite: 37 passing unit & integration tests executed via pytest with automated coverage across all layers.

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SYSTEM BOUNDARIES & ROADMAP

Current Limitations & Production Improvements

Current Technical Limitations
• Scan Quality Sensitivity: Highly degraded, blurred, crumpled, or low-DPI scans can impact
decimal place readability.
• Complex Non-Standard Layouts: Heavily customized multi-column banking footnotes or
unconventional tables can require specialized post-processing.
• LLM Inference Latency & Quotas: Cloud multimodal LLM calls take several seconds per document and require API rate-limit management on free-tier quotas.
• Ephemeral Local Storage: SQLite embedded storage on free-tier containers does not persist across container re-deployments.

Enterprise Production Roadmap
• Managed Relational Database: Transition to AWS RDS / Google Cloud SQL PostgreSQL with JSONB indexing for enterprise durability.
• Durable Object Storage: Store uploaded files and rendered page images in AWS S3 or Google Cloud Storage with presigned access.
• Asynchronous Task Queue: Integrate Celery + Redis for async document processing, decoupling ingestion from AI extraction.
• Security & Access Control: Implement OAuth2 / JWT authentication, role-based access control
(RBAC), and rate limiting.
• Human-in-the-Loop (HITL): Automated routing of low-confidence extractions (< 80%) to human
verification portals.
• Observability: Structured OpenTelemetry distributed tracing, Prometheus metrics, and Sentry error alerting.

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TECHNICAL DEMONSTRATION

Live Demonstration & Evaluator Walkthrough

Step 1 Open Deployed Frontend — Access the web application dashboard in any browser at the public deployment URL.
Step 2 Select Document Type — Choose one of: Invoice, Balance Sheet, Profit & Loss, or Cash Flow Statement.
Step 3 Upload Document — Attach a sample file (e.g. receipt JPG or multi-page financial PDF) and click 'Process'.
Step 4 Inspect Extracted Data — Review extracted key-value fields, confidence badges, and multi-row line item tables.
Step 5 Verify Financial Audits — Examine mathematical validation checks: formulas, calculated vs reported, variances, and PASS/FAIL/NA.
Step 6 Inspect Raw JSON Contract — Click 'Toggle Raw JSON' to review the standardized API payload structure and metadata.
Step 7 Explore Swagger & REST API — Navigate to /docs, execute GET /api/v1/documents, and test GET /api/v1/health.

Evaluation Principle: The system runs transparently in front of evaluators. Upload any provided benchmark file to inspect real-time OCR, extraction, validation, and database indexing.

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SUBMISSION VERIFICATION

Comprehensive Case Study Deliverables Checklist

Core Requirements Verification
✓ Public GitHub Repository: Clean commit history, modular structure.
✓ Live Deployed Frontend: Accessible web dashboard on Render cloud.
✓ Live Backend API: Fully functional RESTful endpoints with CORS.
✓ Interactive Swagger Docs: OpenAPI specification available at /docs.
✓ Health Check Endpoint: Monitoring probe at /api/v1/health.
✓ Comprehensive README: Setup guide, architecture, API docs & trade-offs. ✓ Solution Presentation PDF: Professional 12-slide technical submission.
✓ Sample JSON Outputs: Real extracted outputs in sample_outputs/.

Technical Coverage Verification
✓ All 4 Document Types: Invoice, Balance Sheet, P&L, and Cash Flow.
✓ Scanned Document Support: Tested on scanned banking & receipt images. ✓ Relational Persistence: SQLite storage with SQLAlchemy ORM upsert.
✓ Automated Test Suite: 37 passing pytest unit and integration tests.
✓ Validation Failure Testing: Real-world discrepancy detection tested.
✓ Security & Error Handling: Zero exposed API keys or raw stack traces.
✓ Engineering Discipline: Clean design patterns, no hardcoded extraction. ✓ Production Roadmap: Clear articulation of limits and enterprise scale.

Status: Deployed, testable, documented and ready for technical walkthrough.

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